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Outline to prove Unsolvability by radicals.

Facts

- Radical extensions can be symmetrized.
  i.e. embeded into a new rad. ext. which has the property of symmetry.

  * Explain the connection between symmetrization process and build rad. ext. that extends all permutations σ of $x_1, \dots, x_n$.

- $\mathbb{P}$ polynomial $x^n + a_{n-1}x^{n-1} + \dots + a_1 x + a_0 = 0$.

$\mathbb{P}$ has soltn by radicals

$\Downarrow$ ~~Theorem A~~ entails (how?)

$\exists$ E radical extension of $\mathbb{Q}(a_0, \dots, a_{n-1})$ — It does! (why?)

$\Downarrow$ ~~Theorem A~~ (F1) + E must contain $x_1, \dots, x_n$

E is a radical ext. of $\mathbb{Q}(x_1, \dots, x_n)$

$\Downarrow$ Theorem 1

$\exists$ $\overline{E} \supseteq E$ new rad ext. that has symmetry property — solo se usa para derivar el corolario

$\cdot$ f E

~~Note $\overline{E}$ is rad ext of $\mathbb{Q}(a_0, \dots, a_{n-1})$~~

Corollary 1

Existe new $\overline{E}$ rad ext. of E con prop. del grupo Gal

i.e. $\mathrm{Gal}\left(\dfrac{\overline{E}}{\mathbb{Q}(a_0, \dots, a_{n-1})}\right)$ has property (P)

In summary:

If $\mathcal{P}$ has a solution by radicals $\Longrightarrow$ $\exists\, \overline{E}$ ~~with property~~ r.e. containing $x_1, \dots, x_n$ s.t. $Gal\left(\dfrac{\overline{E}}{\mathbb{Q}(a_0, \dots, a_{n-1})}\right)$ has property (P).

i.e. hemos ~~traducido~~ codificado una proposición ~~en~~ con un objeto algebraico.

Idea: Use the converse to show non existence of sln by radicals.

<u>Enough:</u> Gal group does not have property (P). (for $n \geq 5$)

First Section

§ THE STRUCTURE OF R. Ext.

~~Summary~~ General idea: Prove that Gal group has a property (N) called solvability.

<u>Outline:</u>

- ~~Decompose~~ Construct an ascending tower of fields starting from $\overline{E}$

$\Downarrow$ it gives rise to

a descending tower of ~~fields~~ groups $\boxed{\text{which have property (N)}}$ $\Longleftarrow$ $\boxed{\text{Theorem 2}}$

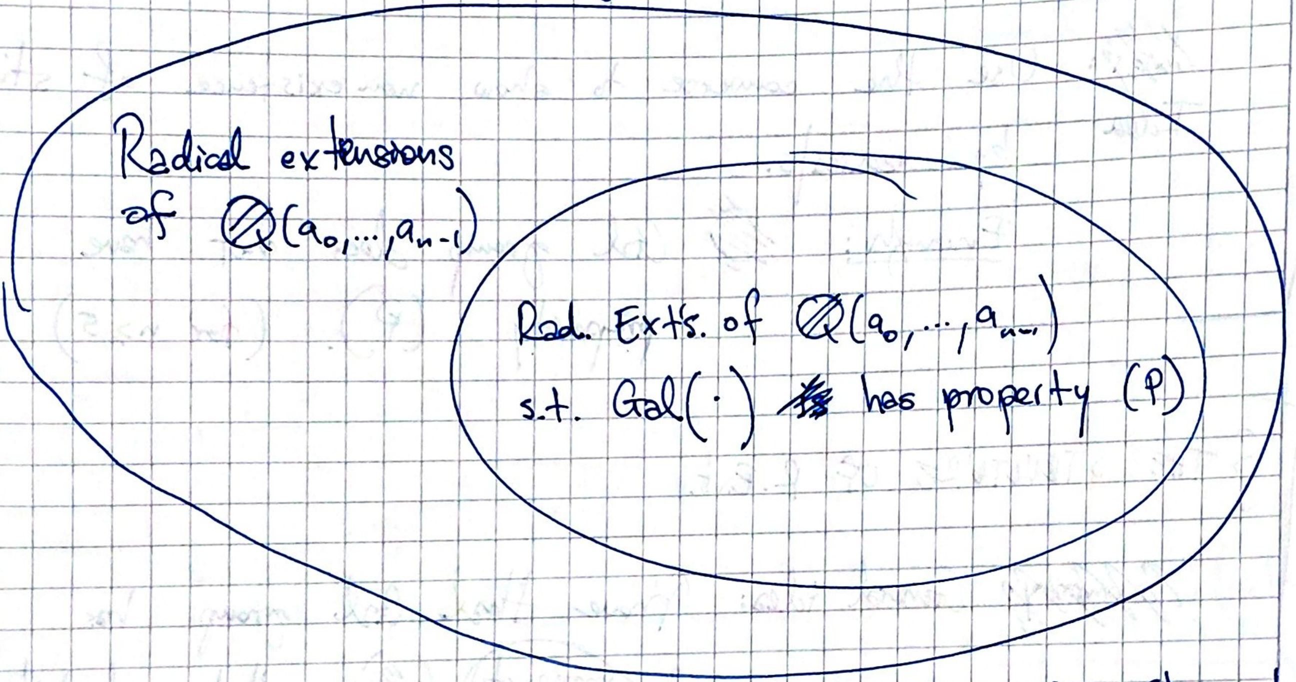
Second Section

§ NON-EXISTENCE OF SOLUTIONS BY RADICALS WHEN $n \geq 5$

<u>GOAL</u> Any rad. ext. of $\mathbb{Q}(a_0, \dots, a_{n-1})$ cannot contain $x_1, \dots, x_n$

(Hence cannot contain $\mathbb{Q}(x_1, \dots, x_n)$)

Thus the object $\overline{E}$ in (*) does not exits



Therefore, appeling to the contrapositive of 1st section, P does not have a solutn by radicals.

How do we prove ~~that~~ the <u>GOAL</u>?

- We use the solvability property (N) given by theorem 2.
- Properties (N) and (P) cannot cohexists.

(N) proves (P) ~~is false~~ does not hold